Convergence rates

Subgradient method : $x^+ = x - \eta g_x$, $g_x \in \partial f(x)$

Gradient descent for smooth functions

Gradient descent for smooth and strongly convex functions

$x^+ = x - \eta \nabla f(x)$

Goal : obtain bounds on sub-optimality of x_t

$\boxed{f(x_t) - f(x^*)}$, where x^* solves: min: $f(x)$
st: $x \in \mathbb{R}^n$

what does this gap look like as a function of t .

Sub gradient method for Lipschitz convex functions

Let f be convex. Assume: $\forall x, \forall g_x \in \partial f(x), \ \|g_x\|_2 \le G$ //

sub-gradient method: $x_{t+1} = x_t - \eta g_t, \quad g_t \in \partial f(x_t)$

$\|x_1 - x^*\|_2 \le R$

$$\underbrace{\|x_{t+1} - x^*\|_2^2} = \|x_t - \eta g_t - x^*\|_2^2$$

$$= \|x_t - x^*\|_2^2 - 2\eta \underbrace{\langle g_t, x_t - x^* \rangle} + \boxed{\eta^2 \|g_t\|_2^2}$$

↳ should be thinking about key def'n of

convexity: $f(y) \ge f(x) + \underbrace{\langle g_x, y - x \rangle}$

$$\le \|x_t - x^*\|_2^2 - 2\eta \underbrace{(f(x_t) - f(x^*))} + \cancel{\eta^2 \|g_t\|_2^2} \ \eta^2 G^2$$

$$\left[f(x_t) - f(x^*) \le \frac{1}{2\eta} \left(\underbrace{\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2} \right) + \frac{\eta}{2} G^2 \right]$$

will telescope once we sum over t

dropping this term respects the inequality

$$\frac{1}{T} \sum_{t=1}^{T} (f(x_t) - f(x^*)) \le \frac{1}{2\eta} \cdot \frac{1}{T} \left(\|x_1 - x^*\|_2^2 - \cancel{\|x_T - x^*\|_2^2} \right) + \frac{\eta}{2} G^2$$

$\vee\!\!|$

$$f\left(\frac{1}{T}\sum x_t\right) - f(x^*) \le \frac{1}{2\eta T} R^2 + \frac{\eta}{2} G^2 \quad \therefore \quad \text{Best } \eta \sim 1/\sqrt{T}.$$

Summary of subgradient method:

1. If we plan to run for T iterations

   best step size $\eta \sim \frac{1}{\sqrt{T}}$

2. Error after T iterations $\sim 1/\sqrt{T}$

   To have error ε need $\sim 1/\varepsilon^2$ iterations.

3. This is good news. (a) sub-grad method produces ε-opt. solution

   (b) "dimension-free"

Gradient descent for smooth convex functions

min: $f(x)$, $f(x)$ is a β-smooth, convex function.

Update: $x_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$ $\quad (\eta = 1/\beta)$.

Recall: $f(x_{t+1}) \leq f(x_t) - \frac{\eta}{2} \|\nabla f(x_t)\|_2^2$ $\quad$ (holds for any $\eta \leq 1/\beta$)

$$\leq f(x^*) + \langle \nabla f(x_t), x_t - x^* \rangle - \frac{\eta}{2} \|\nabla f(x_t)\|_2^2$$

$$= f(x^*) + \frac{1}{2\eta} \left(\|x_t - x^*\|_2^2 - \|x_t - x^* - \eta \nabla f(x_t)\|_2^2 \right)$$

$$f(x_{t+1}) \leq f(x^*) + \frac{1}{2\eta} \left(\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 \right)$$

telescope once we sum

$$f(x_T) - f(x^*) \leq \frac{1}{T} \sum \left(f(x_t) - f(x^*) \right) \leq \frac{1}{T} \cdot \frac{1}{2\eta} \cdot \|x_0 - x^*\|_2^2 = \frac{1}{T} \left(\frac{\beta}{2} \cdot R^2 \right)$$

Suboptimality scales like $\sim 1/T$

For error $\varepsilon \sim 1/\varepsilon$ steps.

Note use of the "self-tuning" property of smooth functions.

Gradient descent for smooth and strongly convex functions

Recall co-coercivity of smooth functions

[If f is β-smooth $\Rightarrow \langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{\beta} \| \nabla f(x) - \nabla f(y) \|_2^2$]

Assume f is α-strongly convex & β-smooth.

$f(x) - \frac{\alpha}{2} \|x\|_2^2$ is convex.

Also this is $(\beta - \alpha)$ smooth

(f is γ-smooth if $\frac{\gamma}{2} \|x\|_2^2 - f(x)$ is convex)

Co-coercivity of $\left(f(x) - \frac{\alpha}{2}\|x\|_2^2\right)$ implies:

$$\underbrace{\langle \nabla f(x) - \nabla f(y), x - y \rangle} \geq \alpha \|x - y\|_2^2 + \frac{1}{\beta - \alpha} \| \nabla f(x) - \nabla f(y) \|_2^2 + \frac{\alpha^2}{\beta - \alpha} \|x - y\|_2^2 - \frac{2\alpha}{\beta - \alpha} \underbrace{\langle \nabla f(x) - \nabla f(y), x - y \rangle}$$

Simplifying, we find:

$$\boxed{\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{\alpha\beta}{\alpha + \beta} \|x - y\|_2^2 + \frac{1}{\alpha + \beta} \| \nabla f(x) - \nabla f(y) \|_2^2}$$

Gradient descent for smooth and strongly convex functions

$\eta = \frac{2}{\alpha + \beta}$, $x_{t+1} = x_t - \eta \nabla f(x_t)$

$$\|x_{t+1} - x^*\|_2^2 = \|x_t - \eta \nabla f(x_t) - x^*\|_2^2$$

$$= \underbrace{\|x_t - x^*\|_2^2} - 2\eta \underbrace{\langle \nabla f(x_t), x_t - x^* \rangle} + \eta^2 \underbrace{\|\nabla f(x_t)\|_2^2}$$

$$\| \quad \langle \nabla f(x_t) - \nabla f(x^*), x_t - x^* \rangle \geq \frac{\alpha\beta}{\alpha+\beta} \|x_t - x^*\|_2^2 + \frac{1}{\alpha+\beta} \|\nabla f(x_t) - \nabla f(x^*)\|_2^2$$

$$\leq \left(1 - 2\eta \frac{\alpha\beta}{\alpha+\beta}\right) \|x_t - x^*\|_2^2 + \left(\eta^2 - 2\eta \frac{1}{\alpha+\beta}\right) \|\nabla f(x_t)\|_2^2$$

since $\eta = \frac{2}{\alpha+\beta}$

R^2

$$\|x_{t+1} - x^*\|_2^2 \leq \left(1 - 2\eta \frac{\alpha\beta}{\alpha+\beta}\right) \|x_t - x^*\|_2^2 \leq \left(1 - 2\eta \frac{\alpha\beta}{\alpha+\beta}\right)^t \cdot \|x_1 - x^*\|_2^2$$

$$\Rightarrow \|x_{t+1} - x^*\|_2 \leq \left(\frac{\beta/\alpha - 1}{\beta/\alpha + 1}\right)^t \|x_1 - x^*\|_2$$

// β/α — κ

condition #

Summary:

f α-strongly convex, β-smooth

suboptimality $\sim$ c^T, $c = \left(\frac{\kappa - 1}{\kappa + 1}\right)^{<1}$ $\kappa = \beta/\alpha$.

For error ε: $\sim$ $\log(1/\varepsilon)$ iterations.

sub-gradient: $1/\sqrt{T}$ i.e. $1/\varepsilon^2$

smooth: $1/T$ i.e. $1/\varepsilon$

s.c. & smooth: c^T i.e., $\log(1/\varepsilon)$